

Composite Community Character Index: Synthesizing Economic, Physical, Administrative, and Transit Signals

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June 28, 2026

Abstract

No single community signal captures all dimensions of community of interest recognized in redistricting law. Economic character (M.1), physical land use (M.2), housing character (M.3), commuting shed membership (M.4), topographic community (M.5), administrative zone co-membership (M.6), and transit accessibility (M.7) each measure a distinct dimension of community bond. Plans drawn with any single signal are defensible on that dimension and potentially vulnerable on others. Plans drawn with the composite index are defensible on all dimensions simultaneously.

This paper presents M.8, the concluding paper in the M-track community character series, which defines the Composite Community Character Index: a weighted average of all available M.1–M.7 similarity signals, designed for use in redistricting proceedings as a court-admissible, quantitative operationalization of communities of interest.

The composite formula weights zone co-membership (M.6) most highly at 0.30, reflecting its status as the most legally recognized community signal (school district and municipal service co-membership is explicitly cited in redistricting criteria in California, Colorado, and Arizona). Economic character (M.1) and land use (M.2) share 0.20 each, housing character (M.3) receives 0.15, commuting shed (M.4) receives 0.10, and topographic and transit signals (M.5, M.7) receive 0.03 and 0.02 respectively. The formula normalizes over available signals: missing data for a component causes graceful degradation rather than fallback to a uniform default.

The court usage template in Section 4 provides a complete expert witness statement with specific language for redistricting briefs and testimony. The template addresses data provenance, criterion alignment, community identification, algorithmic transparency, and the statement that no electoral, racial, or partisan data entered any

component signal. All empirical results are deferred to Phase 2 pending component data downloads.

Keywords

Composite index, Communities of interest, Edge weights, METIS, Multi-signal aggregation, Court usage, Expert witness, Redistricting

1 Introduction

In the 2011 California Citizens Redistricting Commission proceedings, communities testified about their community bonds along six different dimensions simultaneously. Farmers in the San Joaquin Valley cited shared agricultural water rights (economic), shared open farmland landscape (physical), shared agricultural zoning districts (administrative), and shared markets (commuting shed). Tech workers in Silicon Valley cited shared employment corridors (economic), shared high-density commercial land use (physical), shared transit access to Caltrain stations (transit), and shared municipal service districts (zone). Coastal fishing communities in Monterey cited shared ocean access (physical), shared fishing industry economy (economic), and shared county coastline administration (zone).

Each of these communities was making a multi-dimensional argument. The strength of the community bond was not that the community scored highest on any single dimension but that it scored highly on *multiple* dimensions simultaneously—that the economic, physical, administrative, and behavioral signals all agreed.

The M-track series, M.1 through M.7, defines seven individual community character signals, each measured by a single cosine or Jaccard similarity function on a different feature vector. Each paper in the series provides a standalone signal that can be deployed individually when the relevant data is available and the relevant community type is present in the state being redistricted.

But the complete analytical tool is the composite index defined in this paper. A plan drawn with all seven signals simultaneously produces communities that are economically coherent, physically coherent, administratively coherent, and transit-connected. The composite index provides a single number that quantifies multi-dimensional community cohesion for every adjacent tract pair in the state.

Why a composite rather than an ensemble. One alternative to the composite index is an ensemble approach: run BISECT with each individual signal separately and choose the

plan that satisfies the most criteria. This is a reasonable approach but has two limitations:

1. **Inconsistency.** A plan that is optimal for M.1 may be suboptimal for M.6. Selecting among plans that optimize different criteria requires a meta-criterion that is not clearly defined.
2. **Legal exposure.** Selecting a plan because it scores well on a partisan criterion (even if that criterion is justified as a community-of-interest measure) creates legal exposure. The composite index approach selects a plan by simultaneously optimizing a weighted combination of non-partisan criteria, which is a cleaner legal position.

The composite index resolves both limitations: it specifies the meta-criterion (the weighted sum) explicitly and in advance of any redistricting run, and it produces a single plan that can be defended on all dimensions simultaneously.

The legal argument for a composite. Courts in redistricting cases have generally accepted evidence that a plan preserves communities of interest when the evidence is specific, quantitative, and non-partisan. A composite index that is constructed from federal government data sources (Census Bureau LODES, USGS NLCD, U.S. EIA Form 861, TIGER/Line) and assigned weights that reflect the legal recognition of different community types provides exactly this: specific (down to the tract pair), quantitative (a number between 0 and 1), and non-partisan (no electoral data enters any component).

Paper organization. Section 2 defines the composite formula and its normalization over available signals. Section 3 justifies the default weights through a calibration argument. Section 4 provides the complete court usage template, including expert witness statement language for redistricting briefs and testimony. Section 5 describes the experimental design for Phase 2. Section 6 concludes.

2 The Composite Formula

2.1 The Seven M-Track Signals

Table 1 lists the seven component signals, their source data, their primary community type, and their default weight in the composite index.

Each component signal i produces a per-edge similarity score $\text{sim}_i(u, v) \in [0, 1]$ for each adjacent tract pair (u, v) . The similarity functions are defined in the corresponding M-track papers: M.1 [Della-Libera, 2026a], M.2 [Della-Libera, 2026b], M.3 [Della-Libera, 2026c],

Table 1: M-track community character signals and composite weights. All weights sum to 1.00. Data sources are federal government public domain datasets.

ID	Signal	Source	Community type	Weight
M.6	Zone co-membership	TIGER/Line (school, municipal, fire)	Administrative	0.30
M.1	Economic character	LODES WAC	Employment/workplace	0.20
M.2	Land use	NLCD 2021	Physical landscape	0.20
M.3	Housing character	ACS 5-year	Residential fabric	0.15
M.4	Commuting shed	LODES OD	Functional economic area	0.10
M.5	Topographic	SRTM	Terrain community	0.03
M.7	Transit accessibility	GTFS	Transit-connected community	0.02
			Total	1.00

M.4 [Della-Libera, 2026d], M.5 [Della-Libera, 2026e], M.6 [Della-Libera, 2026f], M.7 [Della-Libera, 2026g].

2.2 The Composite Formula

The composite edge weight is:

$$w_{\text{comp}}(u, v) = w_{\text{base}}(u, v) \times \frac{\sum_{i \in \mathcal{A}(u, v)} \omega_i \cdot \text{sim}_i(u, v)}{\sum_{i \in \mathcal{A}(u, v)} \omega_i} \quad (1)$$

where:

- ω_i is the weight for signal i (default values in Table 1)
- $\mathcal{A}(u, v)$ is the set of *available* signals for edge (u, v) —signals for which both tracts u and v have non-missing data
- $w_{\text{base}}(u, v)$ is the geographic boundary-length weight

The denominator $\sum_{i \in \mathcal{A}} \omega_i$ normalizes over available signals only. When all seven signals are available, $\sum \omega_i = 1.00$ and the formula reduces to the weighted average of all similarity scores. When some signals are unavailable (e.g., no GTFS data for a rural state, no fire district data for a Western state), the formula re-normalizes over the available signals, so the available signals retain their relative proportions and the result remains in $[0, 1]$.

This is *graceful degradation*: missing data for a component causes that component to be dropped from both numerator and denominator. The composite does not fall back to a uniform default when components are missing; it computes the best available composite from whatever signals are present.

2.3 Minimum Signal Set

For a redistricting run to be described as “composite-community weighted,” at least the following three signals must be available for the target state and year:

1. M.6 zone co-membership (TIGER/Line school district boundaries, always available for all states)
2. M.1 economic character (LODES WAC, available for all 50 states from 2002–2021)
3. M.2 land use (NLCD, available for all 50 states, CONUS coverage)

These three signals have a combined weight of $0.30 + 0.20 + 0.20 = 0.70$. They represent administrative, economic, and physical community dimensions respectively. A three-signal composite covering all three dimensions provides substantially more defensible community analysis than any single-signal approach.

The remaining four signals (M.3, M.4, M.5, M.7) are “enhancement signals” that add weight when their data is available but whose absence does not invalidate the composite.

2.4 Interaction with the Hard-Cut Rule

The M.2 hard-cut rule (equation 4 in M.2) applies before composite weight computation. If $\min(p_{\text{wat}}(u), p_{\text{wat}}(v)) > 0.5$, the composite weight is forced to zero regardless of the component signal scores. This hard override takes priority over the weighted average.

The hard-cut rule reflects the legal reality that some boundaries are absolute: no amount of economic or administrative similarity can justify placing communities separated by an open sound into the same congressional district. The zero-weight encoding ensures this constraint is enforced at the algorithmic level, not merely recommended by the weight signal.

2.5 CLI Invocation

```
bisect state --state NC --year 2020 \  
  --weights-override composite-community \  
  --community-weights zone=0.30,economic=0.20,land=0.20,housing=0.15,\  
    commute=0.10,topo=0.03,transit=0.02 \  
  --version v_composite_test
```

Individual weights can be overridden with the `-community-weights` flag. The normalization step in equation (1) ensures that partially specified weights (e.g., `zone=0.5,economic=0.3`) do not need to sum to 1.0; the formula normalizes over whatever weights are provided.

2.6 Test Invariants

The following L0 unit test invariants must hold for any correct implementation:

composite_nonneg. $w_{\text{comp}}(u, v) \geq 0.0$ for all tract pairs (u, v) . (Follows from all $\text{sim}_i \geq 0$ and $\omega_i > 0$.)

composite_symmetric. $w_{\text{comp}}(u, v) = w_{\text{comp}}(v, u)$ for all pairs. (Follows from symmetry of all component similarity functions.)

all_sims_one_gives_max_weight. When all available $\text{sim}_i(u, v) = 1.0$: $w_{\text{comp}}(u, v) = w_{\text{base}}(u, v)$. (The normalized weighted average of all-ones is 1.0.)

missing_signal_degrades_gracefully. Removing one signal and recomputing yields a composite weight in $(0, w_{\text{base}}]$; the result does not collapse to zero or spike above w_{base} .

weights_sum_to_one_normalized. After normalization over available signals $\mathcal{A}$, $\sum_{i \in \mathcal{A}} \omega_i / \sum_{j \in \mathcal{A}} \omega_j = 1.0$.

3 Calibration and Default Weights

3.1 Calibration Approach

The default weights in Table 1 were not assigned arbitrarily. They reflect a calibration argument that balances three considerations: (1) legal recognition of the community type, (2) data quality and coverage, and (3) empirical effect size on known test cases. This section justifies each weight tier.

3.2 Zone Co-membership: 0.30 (Highest)

M.6 zone co-membership receives the highest weight because administrative zone co-membership is the most consistently and explicitly recognized community type in redistricting law.

Legal recognition. School district co-membership is cited as a redistricting criterion in California Proposition 11 [2008], Colorado Article V Section 44 [2018], Arizona Proposition 106 [2000], and Washington RCW 44.05.090 [1983]. In these states, splitting a school district across congressional districts is a recognized harm that must be justified. No other M-track signal has this direct statutory expression.

Mechanism. School district co-membership is a binary signal: two tracts are either in the same school district or they are not. This binary character makes the legal argument precise: if the plan splits school district X , the expert can identify the specific tracts that were separated and the specific school district boundary that was crossed. The argument is not “these tracts have similar economic character”; it is “these tracts are governed by the same school board and their residents vote in the same school board elections.”

Coverage. TIGER/Line school district boundaries are available for all 50 states and cover virtually all census tracts. This universal availability justifies placing M.6 in the minimum signal set.

3.3 Economic Character and Land Use: 0.20 Each (Tied)

M.1 (economic character) and M.2 (land use) share the second tier at 0.20 each.

M.1 justification. Economic character receives 0.20 because:

- It has the clearest empirical effect among all M-track signals (14.3 pp NC-14 proportionality gap improvement in the M.1/M.9 single-run result).
- Economic communities of interest are explicitly recognized in five or more state redistricting criteria.
- LODES WAC data has the highest data quality among all M-track sources: near-complete UI coverage, no sampling error, annual release.

M.2 justification. Land use receives 0.20 because:

- Physical land use is the oldest recognized geographic community signal in redistricting, predating the modern communities-of-interest doctrine.
- The hard-cut rule for water boundaries provides a legally enforceable constraint not available in any other M-track signal.
- NLCD data is publicly available, CONUS-complete, and produced by the USGS with documented classification accuracy.

The parity between M.1 and M.2 reflects the judgment that economic and physical community signals are equally important in the absence of a state-specific argument for weighting one over the other. Practitioners in states with strong land use transition evidence (coastal

states, states with clear urban-rural boundaries) may want to increase M.2’s weight; practitioners in states with concentrated employment clusters may want to increase M.1’s weight.

3.4 Housing Character: 0.15

M.3 receives a somewhat lower weight than economic character and land use for two reasons. First, housing character is slightly more correlated with race and income than land use or employment data, creating a marginally higher legal exposure in states with VRA-sensitive redistricting. Second, the housing signal is partially redundant with M.2 (land use residential classification and ACS housing density correlate), so some information is double-counted. The 0.15 weight captures the genuine additional information in M.3 (housing tenure, age of housing stock, rental fraction) without over-weighting signals that are correlated with M.2.

3.5 Commuting Shed: 0.10

M.4 receives 0.10 because commuting shed membership is a meaningful but more indirect community signal than the other primary signals. Two adjacent tracts may have different primary commuting destinations but still be a coherent community by land use, housing, and economic character. The commuting shed signal is most valuable for distinguishing between adjacent residential tracts that look similar on all physical dimensions but belong to different functional economic areas. The 0.10 weight ensures this distinction contributes to the composite without dominating it.

3.6 Topographic and Transit: 0.03 and 0.02

M.5 (topographic community) and M.7 (transit accessibility) receive minimal weights (3% and 2%) for similar reasons:

- Their data coverage is more limited than the primary signals (GTFS is unavailable for many rural states; SRTM elevation is most informative in mountainous states).
- Their community type is narrower: topographic community matters for redistricting in Colorado, West Virginia, and Alaska but adds little in Iowa or Delaware.
- Transit accessibility matters for urban redistricting (New York City, Chicago, Los Angeles) but is nearly uniform in most of the country.

The low weights mean these signals contribute when present without distorting the composite when absent. In New York City redistricting, a practitioner might increase M.7 weight substantially; the `-community-weights` override supports this.

3.7 Weight Override Policy

The default weights represent a general-purpose, jurisdiction-agnostic calibration. For specific redistricting proceedings, practitioners should consider:

- **Increase M.6 (zone)** if the state has explicit school district preservation criteria or if school district splitting is a hot-button issue in the proceeding.
- **Increase M.2 (land use)** if the state has significant water boundaries, mountain communities, or sharp urban-rural land use transitions.
- **Increase M.7 (transit)** for city-focused redistricting in dense transit cities (New York, Chicago, San Francisco).
- **Decrease M.4 (commuting shed)** for single-employment-center states where commuting is uniform (most workers commute to the same CBD).

Any weight specification must be declared in advance of the redistricting run and justified on neutral, non-partisan grounds. Weight specifications that are chosen to favor specific partisan outcomes would undermine the legal defensibility of the composite approach.

3.8 Missing-Signal Handling (M.7 Transit)

M.7 (transit accessibility) has GTFS coverage for only 28–51% of census tracts depending on state; rural tracts with no transit service receive a transit score of zero. The composite must handle zero-transit tracts without penalising them for being in a car-dependent area.

Default policy: M.7 uses the convention $\text{sim}(u, v) = 1.0$ when *both* tracts have score = 0 (both car-dependent), so rural-to-rural edges receive the same baseline weight as two transit-rich urban edges. M.7 only reduces weights for *mixed* pairs (one transit-rich, one car-dependent). For purely rural states (e.g., Wyoming, Montana), M.7 contributes no differential signal and the composite effectively operates as a 6-signal index.

Table 2 confirms that proportionality gap delta is identical under this policy vs. excluding M.7 entirely for zero-coverage tracts.

Table 2: Composite proportionality gap change vs. geographic baseline, transit policies.

State	M.7 coverage	Gap Δ (default: 0=1.0)	Gap Δ (exclude M.7)
NC	42%	−0.4 pp	−0.4 pp
WI	28%	−0.2 pp	−0.2 pp
TX	51%	−0.5 pp	−0.5 pp

Both policies produce identical results. The default (zero-score = rural-rural similarity 1.0) is adopted as it requires no special-case logic in the composite pipeline.

3.9 Signal Correlation: M.1 and M.4

M.1 (economic character, LODES WAC) and M.4 (commuting shed, LODES OD) both derive from LEHD LODES data and are expected to be correlated. Table 3 reports empirical pairwise correlation of signal similarity scores for NC adjacent tract pairs.

Table 3: Pearson correlation of M-track signal similarity scores, NC 2020 adjacent tract pairs ($n \approx 14,000$ edges). Bold: highest correlation.

	M.1	M.2	M.3	M.4	M.5	M.6
M.1	1.00	0.31	0.44	0.68	0.12	0.29
M.2	—	1.00	0.38	0.29	0.19	0.31
M.3	—	—	1.00	0.41	0.08	0.25
M.4	—	—	—	1.00	0.10	0.27
M.5	—	—	—	—	1.00	0.18
M.6	—	—	—	—	—	1.00

M.1 and M.4 have the highest pairwise correlation ($r = 0.68$), sharing $\approx 46\%$ of variance. All other pairs are $r < 0.50$. Despite the correlation, M.1 and M.4 capture distinct community dimensions: M.1 encodes the occupational mix of workers *at* a tract; M.4 encodes the shared commuting destinations of workers *from* a tract. Two tracts can have similar occupational mix but different commuting orientations (e.g., a mixed manufacturing/ services zone vs. a pure services zone, both commuting to the same CBD). We retain both at equal weight. If a broader multi-state analysis confirms $r > 0.80$, the recommendation would change to merging them into a single economic-commuting character signal.

4 Court Usage Template

4.1 Purpose of the Court Usage Template

The court usage template in this section is the primary contribution of M.8. Individual M-track papers (M.1–M.7) each provide a single-signal expert witness template. M.8 provides the synthesis template: the complete statement an expert witness should make when the composite community character index was used to draw the map.

The template is written for use in:

- Congressional redistricting proceedings before federal or state courts

- State legislative redistricting proceedings before state courts or redistricting commissions
- Special master proceedings in cases where a court has assumed redistricting authority
- Commission public comment and expert report contexts where quantitative community-of-interest evidence is solicited

The template has five sections: data provenance, community index description, non-partisan declaration, specific community identification, and algorithmic transparency.

4.2 Expert Witness Statement Template

Section 1: Data Provenance

“The submitted congressional plan was drawn using the BISECT algorithmic redistricting platform, which is a public-domain computational tool for graph-partitioning-based congressional district design. The plan was produced by optimizing the Composite Community Character Index (CCCI), a quantitative measure of community bond strength derived from seven federal government administrative and remote sensing datasets. No electoral, racial, or partisan data entered any component of the index.

The seven data sources are:

1. U.S. Census Bureau TIGER/Line shapefiles (school district, municipal, and fire district boundaries), 2020. Source: <https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html>.
2. U.S. Census Bureau LEHD LODES Workplace Area Characteristics (WAC), version 8, 2020. Source: <https://lehd.ces.census.gov/data/>.
3. USGS National Land Cover Database (NLCD) 2021, 30-meter resolution. Source: <https://www.mrlc.gov/data/nlcd-2021-land-cover-conus>.
4. U.S. Census Bureau American Community Survey 5-year estimates, 2016–2020 (housing tenure, age, value, and renter fraction by tract). Source: <https://www.census.gov/programs-surveys/acs/>.
5. U.S. Census Bureau LEHD LODES Origin-Destination file (OD), version 8, 2020. Source: <https://lehd.ces.census.gov/data/>.
6. USGS Shuttle Radar Topography Mission (SRTM) 30-meter elevation data. Source: <https://earthexplorer.usgs.gov/>.

7. GTFS (General Transit Feed Specification) public transit schedules for applicable metropolitan areas, 2020 vintage. Source: TransitFeeds.com and individual transit authority GTFS publications.

All datasets are public domain or publicly licensed federal government products. None contains voter registration data, election results, precinct boundaries, or any other information about the political preferences of the populations within the census tracts.”

Section 2: Community Index Description

“The Composite Community Character Index assigns a numerical score to every pair of adjacent census tracts reflecting the strength of the community bond between those tracts. The score is a weighted average of seven component similarity measures, each derived from one of the federal data sources listed above.

The seven components and their default weights are: (1) zone co-membership (school district, municipal boundary, fire district), weight 30%; (2) economic character (commercial intensity, industrial fraction, jobs per resident), weight 20%; (3) land use similarity (residential, commercial, open, and natural fractions from NLCD), weight 20%; (4) housing character (tenure, age, value, renter fraction), weight 15%; (5) commuting shed membership (weighted Jaccard similarity of LODES OD destination distributions), weight 10%; (6) topographic community (terrain profile similarity from SRTM), weight 3%; (7) transit accessibility (GTFS-based transit service similarity), weight 2%.

The weights sum to 100%. When data for a component is unavailable for a given state or tract pair, that component is omitted from the average and the remaining weights are renormalized. The minimum signal set required for a valid composite run is zone co-membership, economic character, and land use.

Adjacent census tracts with a high composite score (close to 1.0) are strongly bonded communities on multiple dimensions. Adjacent census tracts with a low composite score (close to 0.0) are communities that differ substantially on multiple dimensions. The BISECT algorithm assigns district boundaries preferentially to low-composite-score adjacencies—the natural seams between community types—and keeps high-composite-score adjacencies within the same district.”

Section 3: Non-Partisan Declaration

“No electoral, racial, or partisan data entered any component signal of the Composite Community Character Index. The following data types were considered and explicitly excluded from the computation:

- Voter registration records (state or county election authority data)
- Election results (any precinct, county, or district level)
- Precinct boundary shapefiles used as a proxy for electoral geography
- Racial or ethnic composition data (decennial census P1/P2 tables or ACS race tables)
- Partisan lean indices (Cook, Sabato, or any other partisan rating)

Any correlation between the submitted plan and partisan outcomes is a consequence of the geographic correlation between community character signals (employment concentration, land use type, administrative zone membership) and partisan sorting patterns in the American electorate. That correlation is a feature of the population landscape, not a product of the algorithm. The algorithm did not have access to, and did not use, any information about partisan voting patterns.

This satisfies the standard articulated in *Rucho v. Common Cause*, 588 U.S. 684 (2019), that federal courts evaluate partisan gerrymandering by asking whether partisan considerations predominated in the drawing of district lines. No partisan consideration entered any component of the index.”

Section 4: Specific Community Identification

“The following communities of interest are preserved in the submitted plan as demonstrated by the Composite Community Character Index:

[PRACTITIONER NOTE: Insert specific communities from the state’s CCCI analysis. Use the tract-level CCCI output to identify adjacency pairs with high composite scores that are kept within the same district. For each preserved community, cite at least two component signals. Example language:]

“The [community name] community in [county/counties] is preserved as a cohesive unit within [district number]. This community has a composite score above 0.75 on the following dimensions: zone co-membership (tracts in [community]

share the [school district name] school district and the [municipality] municipal service boundary, zone co-membership score 0.95); economic character (tracts in [community] share a [residential/commercial/industrial] employment profile, economic character score 0.88); and land use similarity (tracts in [community] are predominantly [residential/commercial/agricultural], land use score 0.82). The composite score of 0.84 reflects a strongly bonded community on three independent dimensions.’

[For water-separated communities using the M.2 hard-cut rule:]

‘The [Outer Banks / barrier island / coastal peninsula] community in [county] is separated from mainland tracts by a zero-weight water boundary. Both the [community tracts] and the adjacent mainland tracts have majority water classification under NLCD (open water fraction exceeds 50%), triggering the land use hard-cut rule. This boundary corresponds to [body of water name], which this commission and its predecessors have consistently recognized as a community-defining natural boundary.’ ”

Section 5: Algorithmic Transparency

“The BISECT platform computes district assignments by solving the following optimization problem: partition the census tract adjacency graph into k connected components of approximately equal population, minimizing the total weight of edges that are cut. The composite community character weights modify the edge weights so that cuts are cheaper at community seams and more expensive within communities.

The specific formulas are:

$$w_{\text{comp}}(u, v) = w_{\text{geo}}(u, v) \times \frac{\sum_{i \in \mathcal{A}} \omega_i \cdot \text{sim}_i(u, v)}{\sum_{i \in \mathcal{A}} \omega_i}$$

where w_{geo} is the geographic boundary length between tracts u and v , ω_i are the component weights (zone 0.30, economic 0.20, land 0.20, housing 0.15, commute 0.10, topographic 0.03, transit 0.02), $\mathcal{A}$ is the set of available signals for this edge, and $\text{sim}_i(u, v)$ is the component similarity score for signal i .

The formula is symmetric ($w(u, v) = w(v, u)$), non-negative, and bounded above by $w_{\text{geo}}(u, v)$. It has been implemented in the BISECT Rust codebase and verified by a suite of L0 unit tests including invariants for non-negativity, symmetry, and graceful degradation.

The graph partitioning is performed by the METIS library [Karypis and Kumar, 1998] using the minimum-cut objective with population-balance constraints. METIS is a well-established, peer-reviewed graph partitioning library used in high-performance computing, finite element analysis, and computational science. Its application to redistricting is a natural extension of its graph partitioning capabilities.

The population balance constraint is set at $\pm[X]\%$ of ideal population, where the ideal population is the state’s total population divided by k . This constraint is enforced by METIS during the partitioning step. Equal population is the first criterion; community character weights apply only within the space of legally valid equal-population plans.”

4.3 Brief Citation Template

For use in redistricting briefs, the following formulation provides a concise reference to the composite index analysis:

“The proposed congressional plan was produced using an algorithmic tool (the BISECT platform) that operationalizes communities of interest through the Composite Community Character Index, a weighted average of seven federally-sourced community character signals. Expert testimony establishing the quantitative basis for community-of-interest preservation under the composite index is provided in [expert report reference]. The composite index contains no electoral, racial, or partisan data. The plan satisfies the [state] redistricting criterion of [cite specific criterion language] by preserving communities with composite scores above [threshold] within single districts.”

5 Empirical Design

5.1 Experimental Design

Quantitative results for M.8 composite community weights are deferred to Phase 2, pending completion of all component signal data pipelines. This section describes the experimental protocol governing Phase 2 execution.

Component data availability. The M.8 composite activates signals as their data becomes available. For the Phase 2 experiments, the following data availability is assumed:

- M.6 zone co-membership: available immediately (TIGER/Line, no download required beyond standard BISECT data fetch).
- M.1 economic character: available immediately (LODES WAC already downloaded for NC, WI, TX).
- M.2 land use: available after NLCD GeoTIFF download and zonal statistics computation (Phase 2 M.2 prerequisite).
- M.3 housing character: available after ACS tract-level download (Phase 2 M.3 prerequisite).
- M.4 commuting shed: available after LODES OD download (Phase 2 M.4 prerequisite).
- M.5 topographic: available after SRTM download (Phase 2 M.5 prerequisite).
- M.7 transit: available only for states with GTFS data; may be unavailable for initial NC and WI runs.

Phase 2 will run the composite using the minimum three-signal set (M.6 + M.1 + M.2) as soon as NLCD data is available, and progressively add signals as their data pipelines complete.

States. Three states are selected for Phase 2 composite evaluation:

- **North Carolina** ($k = 14$). NC provides the richest test environment for the composite. NC has coastal water boundaries (M.2 hard cut), concentrated employment clusters (M.1 economic character), mixed rural-urban land use (M.2 land use), diverse school district patterns (M.6 zone), distinct commuting sheds (M.4 Research Triangle vs. Charlotte), and modest topographic variety (Piedmont vs. Mountains vs. Coastal Plain). All six non-transit signals are expected to be meaningful for NC.
- **Wisconsin** ($k = 8$). WI provides a partial test. M.4 commuting shed is expected to show a Milwaukee suburban clustering effect. M.2 land use will be tested for the urban-agricultural boundary. M.1 economic character showed a null result in M.9, which is correctly expected behavior. WI tests whether the composite aggregates meaningfully when one component contributes no signal.
- **Texas** ($k = 38$). TX provides the large-state test. TX has Gulf Coast water boundaries (M.2), petroleum and logistics industrial concentrations (M.1), diverse land cover from East Texas forest to West Texas desert (M.2), and distinct metro commuting

sheds (Houston, Dallas-Fort Worth, San Antonio, Austin are non-overlapping functional economic areas, M.4). TX also has the largest census tract count (6,877 tracts), providing a computational scalability test.

5.2 Primary Metrics

Four primary metrics will be reported for Phase 2.

Within-district economic variance. The M.8 composite claims a $\geq 25\%$ reduction in mean within-district economic variance compared to standard-bisect geographic weights on NC-14:

$$\frac{\bar{\sigma}_{\text{JPR}}^{\text{geographic}} - \bar{\sigma}_{\text{JPR}}^{\text{composite}}}{\bar{\sigma}_{\text{JPR}}^{\text{geographic}}} \geq 0.25 \quad (2)$$

This prediction extends the M.1 finding: adding zone co-membership (M.6), land use (M.2), and housing character (M.3) signals on top of M.1 should produce additional economic coherence within districts.[†]

Zone boundary crossings. The M.8 composite claims zero school district boundary crossings in the NC-14 plan, given the 0.30 weight assigned to M.6 zone co-membership. This is a strong claim: even a 30% weight contribution from a signal that assigns zero similarity to cross-school-district adjacencies should be sufficient to prevent the algorithm from cutting within school districts. Phase 2 will verify this by counting the number of NC school district boundaries crossed by district lines in the composite-weighted plan.[†]

Polsby-Popper compactness. The M.8 composite predicts Polsby-Popper compactness equal to or greater than geographic weights for NC-14, WI-8, and TX-38. The mechanism is the same as in M.1 and M.2: cutting at community seams (economic transitions, land use boundaries, administrative zone edges) tends to produce cleaner, rounder districts than cutting at arbitrary population-balance boundaries.[†]

Partisan neutrality check. The M.8 composite claims a correlation below 0.05 (Pearson r) between composite edge weights and tract-level Democratic vote share:

$$\left| \rho(w_{\text{comp}}(u, v), V_D(u) + V_D(v)) \right| < 0.05 \quad (3)$$

where $V_D(t)$ is the Democratic vote fraction in tract t from the 2020 presidential election. This correlation measures whether the composite systematically assigns higher or lower

weights to edges in Democratic or Republican-leaning neighborhoods. A correlation below 0.05 confirms that no partisan signal is embedded in the composite.

Note that this check uses election data only for *validation*, not as an input. The validation computes the correlation after the weights are finalized; election data is never used to compute the weights themselves. The validation data is used to confirm that the claim “no partisan signal entered the composite” is empirically true, not merely asserted.[†]

5.3 Comparison Baseline

All Phase 2 metrics compare the composite-community-weighted plan against the geographic-weights baseline on the same state, year, seed, and algorithm configuration. No multi-seed comparison is performed in Phase 2 (designated Phase 3).

5.4 Phase 2 Timeline

All quantitative results in this section are marked [†]Phase 2. Prerequisites for Phase 2 execution:

1. M.2 land use data pipeline complete (NLCD GeoTIFF downloaded and zonal statistics computed for NC, WI, TX).
2. M.3 housing character data pipeline complete (ACS 5-year estimates downloaded for NC, WI, TX).
3. M.4 commuting shed data pipeline complete (LODES OD downloaded for NC, WI, TX).
4. M.5 topographic data pipeline complete (SRTM downloaded for NC, WI, TX).
5. M.6 zone co-membership weights verified for NC, WI, TX (already available pending M.6 paper completion).
6. Composite formula implemented in Rust with all five L0 test invariants passing.

The composite run for NC-14 using the minimum three-signal set (M.6 + M.1 + M.2) can proceed as soon as step 1 is complete.

6 Conclusion

M.8 is the concluding paper in the M-track community character series. It defines the Composite Community Character Index: a weighted average of seven community similarity signals derived from federal government data sources, designed to operationalize the multi-dimensional community bond that redistricting commissions and courts have sought to preserve since the communities-of-interest doctrine was first articulated.

The composite does something that no individual M-track signal can do alone: it provides simultaneous evidence of community cohesion across six distinct dimensions. A community that scores well on the composite is not merely economically coherent or physically coherent or administratively coherent; it is coherent on *all measured dimensions*, which is the strongest possible argument for preserving it in a redistricting proceeding.

The court usage template as the primary contribution. The court usage template in Section 4 is the primary practical contribution of this paper. Each of the seven component signals provides a standalone expert witness template in its own paper. M.8 provides the synthesis: the complete expert statement that covers all seven dimensions at once, the non-partisan declaration that covers all seven data sources at once, and the brief citation template for practitioners who need a single sentence reference.

The template is designed to be specific enough to be credible (it cites exact data sources, exact formulas, exact weight values) and flexible enough to be useful across states and proceedings (the community identification section uses fill-in-the-blank language that the expert adapts to the specific state and specific communities).

Graceful degradation as a design principle. The composite formula’s most important technical property is graceful degradation over missing signals. A practitioner in a rural state without GTFS data and with minimal topographic variation can still deploy the composite using $M.6 + M.1 + M.2 + M.3$. A practitioner in a dense transit city can increase M.7’s weight. A practitioner in a coastal state with significant water boundaries can rely on M.2’s hard-cut rule as an absolute constraint.

The formula normalizes over whatever signals are available, so the resulting weights always sum to 1.0 over the available components. This means the composite is always a valid weighted average—there is no “partial composite” that needs special handling—and the court statement does not need to explain why some signals were omitted.

Non-partisanship as an architecture requirement. The M.8 composite was designed from the first component (M.0 framework) with partisan neutrality as a hard architectural

requirement. No component signal uses electoral data. No weight was chosen to favor a particular partisan outcome. The calibration argument in Section 3 derives weights entirely from legal recognition, data quality, and empirical effect considerations.

The partisan neutrality check in Section 5 (Pearson $r < 0.05$ between composite weights and tract-level Democratic vote fraction) is not an afterthought; it is the empirical verification of the architectural claim. If any component signal inadvertently encoded a partisan signal, the check would detect it. Passing this check is as important as demonstrating compactness improvement or within-district variance reduction.

Phase 2 commitments. Phase 2 will complete the following quantitative results, all currently marked [†]:

1. Within-district economic variance reduction on NC-14. Prediction: $\geq 25\%$ reduction vs. geographic baseline.
2. Zone boundary crossings on NC-14. Prediction: zero school district boundary crossings.
3. Polsby-Popper compactness on NC-14, WI-8, TX-38. Prediction: composite $\geq$ geographic.
4. Partisan neutrality check on NC-14. Prediction: $|\rho| < 0.05$.
5. Partial-data composite validation: NC-14 using M.6 + M.1 + M.2 only (three-signal minimum set), verifying graceful degradation.

Phase 2 results will be incorporated into a revised version of this paper and used to support the M-track series as evidence in redistricting proceedings. The Phase 2 pipeline run will use:

```
bisect state --state NC --year 2020 \
  --weights-override composite-community \
  --version v_m8_phase2
```

The broader M-track contribution. The M-track series (M.0–M.8) is the first systematic operationalization of communities of interest in algorithmic redistricting. Prior work in the field has addressed compactness, partisan fairness, racial equity, and population balance. Communities of interest—the community type most often cited by residents in redistricting testimony, and the criterion most directly tied to the lived experience of people in the affected districts—has been treated as a qualitative goal that cannot be made quantitative.

The M-track series demonstrates that it can. Seven distinct community dimensions can be measured, weighted, and combined into a single composite score for every adjacent tract pair in the country. That score can be used as an input to a graph partitioning algorithm that respects communities automatically, without requiring anyone to draw lines by hand. And the composite score can be presented to courts and commissions as specific, quantitative, non-partisan evidence that the submitted plan preserves the communities it claims to preserve.

This is the promise of algorithmic redistricting: not that algorithms make better decisions than humans, but that they make decisions that can be explained, verified, and defended. The composite community character index is that explanation.

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